Route formulation with 1+1 protection

Each demand is carried on two paths, a working path and a backup path, that share no fiber, so that a single fiber cut cannot interrupt both. Each path uses the same wavelength on all its links. The model first serves as many demands as possible, then uses the fewest wavelengths.

Data

c	a demand from s_c to t_c, requesting v_c wavelengths
n	a candidate pair of demand c: a working path $W_{c,n}$ and a backup path $B_{c,n}$ with no common fiber (computed before solving, as in the greedy heuristic, at most 3 per demand)
$l_{c,n}$	number of links of $W_{c,n}$ plus number of links of $B_{c,n}$
$R(l,k)$	the candidate paths that use the link (l,k)
$F_{l,k}$	number of fibers of the link (l,k)
λ	a wavelength, $\lambda_0, \ldots, \lambda_{W-1}$ (W wavelengths per fiber)
M	a large number, the cost of not serving a demand

Variables (binary)

$x_{c,n}$	= 1 if demand c is served on pair n
y_c	= 1 if demand c is not served (blocked)
$r^W_{c,n,\lambda}$	= 1 if the working path of pair n uses wavelength λ
$r^B_{c,n,\lambda}$	= 1 if the backup path of pair n uses wavelength λ

Model

$$\min \quad M \sum_c y_c \; + \; \sum_c \sum_n v_c \, l_{c,n} \, x_{c,n}$$

Not serving a demand costs M; each wavelength used on a link costs 1.

$$\sum_n x_{c,n} \; + \; y_c \; = \; 1 \qquad \forall \, c$$

Each demand is served on one pair, or it is blocked.

$$\sum_\lambda r^W_{c,n,\lambda} \; = \; v_c \, x_{c,n} \qquad \forall \, c, \; n$$

The working path of the chosen pair gets v_c wavelengths; a pair that is not chosen gets none.

$$\sum_\lambda r^B_{c,n,\lambda} \; = \; v_c \, x_{c,n} \qquad \forall \, c, \; n$$

The same for the backup path.

$$\sum_{W_{c,n} \in R(l,k)} r^W_{c,n,\lambda} \; + \sum_{B_{c,n} \in R(l,k)} r^B_{c,n,\lambda} \; \le \; F_{l,k} \qquad \forall \, (l,k), \; \lambda \in \{\lambda_0, \; \ldots \, , \lambda_{W-1}\}$$

On each link, each wavelength is used at most once, counting working and backup paths together.

$$x_{c,n}, \quad y_c, \quad r^W_{c,n,\lambda}, \quad r^B_{c,n,\lambda} \; \in \; \{0, 1\}$$

No constraint is needed to keep the working and backup paths apart: every candidate pair is already made of two paths with no common fiber.